Mojang Studios
Cybersecurity Risk Assessment

Dhruv Guda | Portfolio revision: October 21, 2026

Purpose and scope

An educational assessment of 20 candidate assets and risk areas using public information and explicit assumptions. Mojang develops Minecraft within Microsoft following the announced 2014 acquisition [1]. The assessment considers devices, identities, source code, customer data, Realms, Marketplace, distribution, third parties, and monitoring.

No internal audit or penetration test is claimed. Actual architecture, ownership, control coverage, budgets, and incident frequencies were not verified. Proposed controls and modeled scores do not demonstrate deployed safeguards or measured risk reduction.

Method and assumptions

NIST SP 800-30 Rev. 1 informs the assessment concepts [27]. This project defines its own 4 x 4 matrix, score bands, target intervals, and assumed 12-month planning horizon. Scores are qualitative scenario judgments, not calibrated probabilities or expected losses.

Dimension	1	2	3	4
Likelihood	Rare	Unlikely	Possible	Probable
Impact	Negligible	Marginal	Critical	Catastrophic

Score = assumed likelihood x assumed impact. Bands: 1-3 Low; 4-7 Moderate; 8-11 High; 12-16 Very High. Numeric target intervals are separate goals and may cross bands. The editable workbook provides definitions and limitations.

Original impact proxies range from below $10,000 (1), $10,000 to under $100,000 (2), $100,000 to under $1 million (3), and $1 million or more (4). These are illustrative, not company loss estimates; privacy and availability harm also matter.

Cost entries remain $10K/$100K/$1M annual order-of-magnitude placeholders. No quotes or resource calculations support a budget; shared controls can overlap. Owner roles are proposals, not verified appointments. Retained residual scores assume effective controls; impact reductions require further justification.

Decision summary

IDs 4, 5, 6, 11, 13, and 15 exceed their numeric targets. Proposed next steps are further treatment, documented acceptance by an authorized owner, or reassessment. No scores were reduced merely to meet targets.

Log4Shell and third-party mod malware illustrate patching and ecosystem exposure [16, 17]. Neither establishes a Mojang internal breach. Microsoft's March 2024 update supersedes the earlier incident snapshot regarding some source/internal-system access [26]; this parent-company incident is not evidence of a Mojang breach.

Risk-score summary

L and I are assumed baseline scores. Residual is modeled after proposed treatment. All control effectiveness remains unverified.

ID	Asset / risk area	L	I	Inherent	Residual	Target	Status
1	Company provided phones	4	3	12	4	1-4	Within target*
2	Company provided badges	4	4	16	6	4-7	Within target*
3	Storage containing company intellectual property	4	4	16	6	4-7	Within target*
4	Minecraft source code and game builds	3	4	12	8	4-7	Above target
5	Microsoft account authentication and identity data	3	4	12	8	4-7	Above target
6	Customer PII and purchase records	3	4	12	8	4-7	Above target
7	Minecraft Realms worlds and backups	3	4	12	6	4-7	Within target*
8	Realms hosting infrastructure and databases	2	4	8	3	1-4	Within target*
9	Minecraft Marketplace and Minecoin transaction records	3	4	12	6	4-7	Within target*
10	Creator partner content and submission pipeline	3	4	12	6	4-7	Within target*
11	Game update and software distribution systems	3	4	12	8	4-7	Above target
12	Mojang employee accounts and collaboration systems	4	4	16	6	4-7	Within target*
13	Development workstations and code-signing credentials	4	4	16	8	4-7	Above target
14	Minecraft brand, trademarks, and licensing agreements	4	3	12	4	1-4	Within target*
15	Third-party cloud and platform services	4	4	16	9	4-7	Above target
16	Player chat, moderation, and trust-and-safety data	4	3	12	4	1-4	Within target*
17	Production monitoring, logs, and incident-response records	4	4	16	6	4-7	Within target*
18	Physical offices and on-site network equipment	3	4	12	6	4-7	Within target*
19	Customer-support tickets and account-recovery records	3	3	9	4	1-4	Within target*
20	Mojang public websites, APIs, and DNS records	4	3	12	4	1-4	Within target*

*Within the proposed numeric target does not mean accepted or demonstrated safe.

Proposed controls and validation

All roles, costs, safeguards, and validation steps below are proposed. No implementation or test against company systems occurred.

1. Company provided phones

Proposed role: Device-management role | Annual placeholder: $10K

Encrypt managed devices; enforce strong local unlock and updates; revoke account/device access after loss. Biometrics and passcodes may be alternative unlock methods.

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Review device inventory, encryption status, revocation timing, and update coverage.

2. Company provided badges

Proposed role: Physical-security role | Annual placeholder: $10K

Issue individual badges; promptly revoke lost badges; review access when roles change; maintain access logs. Assess necessity and privacy before adding biometrics.

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Test badge revocation and review access logs without exposing employee details.

3. Storage containing company intellectual property

Proposed role: IT/security role | Annual placeholder: $10K

Encrypt storage containing intellectual property; restrict access; protect expected integrity values or use authenticated integrity/signatures; inventory devices and media.

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Verify access restrictions, recovery procedures, and the trust source for integrity checks.

4. Minecraft source code and game builds

Proposed role: Engineering-security role | Annual placeholder: $100K

Use identity-based access with MFA, least privilege, managed devices, protected branches, reviewed changes, and separate release approval. Network controls are an additional layer, not proof of identity.

Next decision / verification: Above target: assess further treatment or obtain documented owner acceptance. Review repository permissions, branch protection, release approvals, and unauthorized-change tests.

5. Microsoft account authentication and identity data

Proposed role: Identity-service owner (assumed) | Annual placeholder: $100K

Use managed identity and phishing-resistant MFA where supported; protect tokens and recovery workflows. If a service stores password verifiers, use modern salted password hashing. Provider ownership of Microsoft-account credentials is not verified.

Next decision / verification: Above target: assess further treatment or obtain documented owner acceptance. Confirm service boundaries and credential ownership; review MFA, recovery controls, and access logs.

6. Customer PII and purchase records

Proposed role: Privacy/data-security role | Annual placeholder: $100K

Classify and minimize personal and purchase data; encrypt recoverable sensitive data in transit/at rest; apply least privilege, retention rules, and access monitoring. Hash passwords separately where stored.

Next decision / verification: Above target: assess further treatment or obtain documented owner acceptance. Review data inventory, access permissions, retention enforcement, and key management.

7. Minecraft Realms worlds and backups

Proposed role: Realms operations role | Annual placeholder: $100K

Set recovery-point and recovery-time objectives based on business needs; protect backup copies; restrict operational access by role; test restoration and monitor unauthorized changes.

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Document backup frequency/retention against recovery objectives and record authorized restoration tests.

8. Realms hosting infrastructure and databases

Proposed role: Platform-operations role | Annual placeholder: $1M

1. Continuously patch and harden hosting servers and databases
2. Limit production access with multi-factor authentication and least-privilege permissions
3. Maintain replicated backups and regularly test failover and data recovery

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Test recovery and failover against documented objectives.

9. Minecraft Marketplace and Minecoin transaction records

Proposed role: Commerce-security role | Annual placeholder: $100K

1. Encrypt Marketplace and Minecoin transaction records while stored and transmitted
2. Limit payment and Marketplace administrator access with multi-factor authentication
3. Continuously monitor transaction logs for fraud and unusual account activity

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Review fraud alerts, transaction integrity, and administrator access.

10. Creator partner content and submission pipeline

Proposed role: Creator-platform role | Annual placeholder: $100K

1. Scan all creator submissions for malware and prohibited code before publication
2. Test creator content inside an isolated environment before approving the submission
3. Require verified creator accounts, version history, and approval before content is released

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Test submission scanning, isolation, and approval controls; scanning cannot guarantee safety.

11. Game update and software distribution systems

Proposed role: Release-engineering role | Annual placeholder: $100K

1. Digitally sign every game update and verify file hashes before distribution
2. Limit build and release permissions with multi-factor authentication and separate administrator accounts
3. Monitor distribution systems and maintain a tested rollback copy of the previous stable release

Next decision / verification: Above target: assess further treatment or obtain documented owner acceptance. Verify release signatures against trusted keys and test rollback.

12. Mojang employee accounts and collaboration systems

Proposed role: Identity/security role | Annual placeholder: $10K

1. Require multi-factor authentication and unique passwords for employee accounts
2. Limit account permissions and regularly disable unused or former employee accounts
3. Train employees against phishing and alert security teams about unusual login activity

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Review MFA coverage and departed-user access revocation.

13. Development workstations and code-signing credentials

Proposed role: Engineering-security role | Annual placeholder: $100K

1. Encrypt development workstations and continuously patch and monitor them with endpoint protection
2. Store code-signing credentials in protected hardware with access limited to approved employees
3. Require approval for release signing and keep an audit log of every use of the credentials

Next decision / verification: Above target: assess further treatment or obtain documented owner acceptance. Review protected signing-key use, approval records, and key-revocation procedures.

14. Minecraft brand, trademarks, and licensing agreements

Proposed role: Legal/IP role | Annual placeholder: $10K

1. Store licensing agreements and trademark records in an encrypted centralized repository
2. Limit editing permissions and keep version history for all legal documents
3. Monitor for fraudulent domains, counterfeit content, and unauthorized use of the Minecraft brand

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Review legal-record access and procedures for suspected brand abuse.

15. Third-party cloud and platform services

Proposed role: Vendor-risk role | Annual placeholder: $100K

1. Perform security reviews of third-party vendors before granting access to Mojang systems
2. Require vendors to report vulnerabilities and incidents and to maintain supported, patched services
3. Maintain backups, alternate service options, and a documented plan for vendor outages

Next decision / verification: Above target: assess further treatment or obtain documented owner acceptance. Review vendor evidence, concentration risk, incident obligations, and exit/recovery exercises.

16. Player chat, moderation, and trust-and-safety data

Proposed role: Trust-and-safety role | Annual placeholder: $1M

1. Encrypt chat and moderation records and restrict access to authorized trust-and-safety employees
2. Use automated filtering with human review and an easy process for players to report harmful content
3. Set retention limits and remove or anonymize player data when it is no longer needed

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Review authorized access, retention, escalation, and privacy safeguards.

17. Production monitoring, logs, and incident-response records

Proposed role: Security-operations role | Annual placeholder: $100K

1. Store security logs in a centralized system that restricts changes and preserves evidence of alteration
2. Restrict log access with multi-factor authentication and least-privilege permissions
3. Back up important logs and regularly test incident-response procedures using recorded events

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Test log protection, collection coverage, alert handling, and incident exercises.

18. Physical offices and on-site network equipment

Proposed role: Facilities/network role | Annual placeholder: $100K

1. Require badge access, visitor records, and camera monitoring for restricted office areas
2. Segment office networks and keep firewalls, wireless systems, and network equipment patched
3. Maintain an equipment inventory and lock servers and critical network devices in secure rooms

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Test visitor/access procedures and network segmentation.

19. Customer-support tickets and account-recovery records

Proposed role: Customer-support security role | Annual placeholder: $100K

1. Encrypt customer-support tickets and remove unnecessary payment or identity information
2. Verify a player's identity before account recovery and require multi-factor authentication for support employees
3. Limit ticket access, log employee activity, and delete records after a certain retention period

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Review recovery verification, ticket access, and retention enforcement.

20. Mojang public websites, APIs, and DNS records

Proposed role: Web/platform security role | Annual placeholder: $100K

1. Protect public websites and APIs with a web application firewall, rate limiting, and DDoS protection
2. Secure DNS and domain registrar accounts with multi-factor authentication and registry locks
3. Patch and scan public systems while continuously monitoring certificates, changes, and service availability

Next decision / verification: Within proposed numeric target; effectiveness still unverified. Review rate limits, registrar protection, patching, and availability monitoring.

References and research provenance

Source descriptions and URLs retain the original research trail, with corrected links and added methodology/control references. Historical statements are dated context; not every third-party page was independently revalidated. Community posts are anecdotal leads rather than confirmed company findings. The companion company-research CSV preserves the fuller research notes.

[1] Microsoft News Center - Company identity

Mojang AB (Mojang Studios) is the Stockholm-based developer of Minecraft. Microsoft announced its $2.5 billion acquisition of Mojang in 2014, and Mojang joined Microsoft Studios (now Xbox Game Studios).

https://news.microsoft.com/source/2014/09/15/minecraft-to-join-microsoft/

[2] Microsoft Services Agreement - Business model

Video-game and entertainment business. Revenue paths include sales and licensing of games and digital content, subscriptions, in-game/Marketplace purchases, and distribution through the wider Xbox ecosystem.

https://www.microsoft.com/en/servicesagreement

[3] Minecraft Realms - Business model

Minecraft Realms is a recurring hosted personal-server subscription. Plan prices vary by edition, region, and date; current prices are not asserted in this assessment.

https://www.minecraft.net/en-us/realms

[4] Minecraft Partner Program - Business model

Minecraft Marketplace sells creator-made digital content through Minecoins. Mojang also operates a creator partner program, and Marketplace Pass/Realms Plus provide subscription access to rotating content.

https://www.minecraft.net/en-us/partner

[5] Microsoft acquisition announcement - Critical assets

Minecraft intellectual property: trademarks, game code, art, world formats, proprietary development assets, and the Minecraft brand/franchise.

https://news.microsoft.com/source/2014/09/15/minecraft-to-join-microsoft/

[6] Microsoft Privacy Statement - Critical assets

Player and customer data: names/contact details, email addresses, account credentials, payment data, subscription/license entitlements, purchase history, IP addresses, device identifiers, and connectivity/configuration data.

https://www.microsoft.com/en-us/privacy/privacystatement

[7] Minecraft account guide - Critical assets

Account identity and access data: Microsoft account credentials, Xbox gamertags, purchase entitlements, multiplayer permissions, and account security settings.

https://www.minecraft.net/en-us/article/how-create-minecraft-account

[8] Minecraft Help - Realms backups - Critical assets

Player-created content and service data: Realm worlds, saves, backups, in-game content, and social/usage data associated with online play.

https://help.minecraft.net/hc/en-us/articles/28717462139149

[9] AWS News Blog - Hosting Minecraft Realms on AWS - Critical assets

Candidate assets include authentication, hosting, storage, networking, monitoring, and recovery systems. The linked 2014 deployment is historical evidence, not proof of current architecture.

https://aws.amazon.com/blogs/aws/hosting-minecraft-realms-on-aws/

[10] Minecraft Partner Program - Critical assets

Marketplace creator content, partner submissions, Minecoin transactions, and the trust/safety processes used to review published content.

https://www.minecraft.net/en-us/partner

[11] Mojang Studios LinkedIn company page - Operations

Historical public company-profile research described a distributed workforce and multiple locations. This is not a verified current office inventory.

https://www.linkedin.com/company/mojangstudios

[12] Minecraft Realms - Operations

Minecraft and Realms operate across PCs, consoles, and mobile devices. Bedrock supports cross-platform play across Windows, Android, iOS, Xbox, PlayStation, Nintendo, Amazon Fire, and Chromebook; Java supports Windows, macOS, and Linux.

https://www.minecraft.net/en-us/realms

[13] AWS News Blog - Hosting Minecraft Realms on AWS - Operations

A publicly documented 2014 Realms deployment used AWS EC2, EBS, RDS, S3, Elastic Load Balancing, and multiple AWS Regions. Worlds and game state were stored in S3 at that time.

https://aws.amazon.com/blogs/aws/hosting-minecraft-realms-on-aws/

[14] Guinness World Records - Scale

Minecraft had sold more than 350 million units as of April 2025, according to Guinness World Records, which says the figure was confirmed by Mojang in the official Minecraft Annual 2026.

https://www.guinnessworldrecords.com/world-records/best-selling-video-game

[15] Microsoft FY2025 Q4 earnings call - Scale

Microsoft reported that Minecraft reached record monthly active usage and revenue in FY2025 Q4, but it did not publish a current Minecraft-specific monthly-active-user number in that disclosure.

https://www.microsoft.com/en-us/investor/events/fy-2025/earnings-fy-2025-q4

[16] Microsoft Security Response Center - Recent incident

Log4Shell / CVE-2021-44228 affected Minecraft: Java Edition in December 2021. Mojang patched official clients and Microsoft released a fix blocking the issue in Java Edition 1.18.1; self-hosted server operators also had to update or apply mitigations.

https://www.microsoft.com/en-us/msrc/blog/2021/12/microsofts-response-to-cve-2021-44228-apache-log4j2

[17] CurseForge Support - Recent incident

In June 2023, malware was distributed through infected Minecraft mods/projects on CurseForge and BukkitDev. This was a third-party mod ecosystem incident, not evidence that Mojang's own network was breached.

https://support.curseforge.com/support/solutions/articles/9000228509-june-2023-infected-mods-detection-tool

[18] BleepingComputer - Recent incident

Independent security reporting described the 2023 Fractureiser malware as an information stealer spread through compromised Minecraft mods and plugins.

https://www.bleepingcomputer.com/news/security/new-fractureiser-malware-used-curseforge-minecraft-mods-to-infect-windows-linux/

[19] Microsoft Security Response Center - Recent incident

In January 2024, parent company Microsoft disclosed that Midnight Blizzard accessed a small percentage of corporate email accounts after password-spraying a legacy test tenant. Microsoft stated there was no evidence of access to customer environments, production systems, source code, or AI systems.

https://www.microsoft.com/en-us/msrc/blog/2024/01/microsoft-actions-following-attack-by-nation-state-actor-midnight-blizzard

[20] Minecraft official site - Official website

Minecraft's official website is Minecraft.net. Mojang's official careers page is hosted within Minecraft.net and identifies Mojang AB / Microsoft branding.

https://www.minecraft.net/en-us

[21] Microsoft 2025 Annual Report - Financial filings

Mojang Studios is not separately publicly traded and does not publish its own U.S. Form 10-K. Use parent company Microsoft Corporation's FY2025 Form 10-K for public financial and enterprise-risk disclosures; Microsoft reports gaming within its consolidated business.

https://www.microsoft.com/investor/reports/ar25/index.html

[22] SpigotMC community forum - Industry forum

SpigotMC's Systems Administration forum contains practitioner discussion of Minecraft server security and Log4j.

https://www.spigotmc.org/threads/minecraft-security-bonus-part-1-log4j.538069/

[23] Microsoft Q&A - Industry forum

Microsoft Q&A contains public discussions of compromised Microsoft/Minecraft accounts and recovery problems.

https://learn.microsoft.com/en-us/answers/questions/4103894/minecraft-account-hacked-and-email-taken-off-micro

[24] CISA Joint Cybersecurity Advisory AA21-356A - Cybersecurity report

CISA, FBI, NSA, and international partners published mitigation guidance for Log4Shell and related Log4j vulnerabilities.

https://www.cisa.gov/news-events/cybersecurity-advisories/aa21-356a

[25] NIST NVD - CVE-2021-44228 - Cybersecurity report

NIST's National Vulnerability Database entry for CVE-2021-44228 documents the affected Log4j versions and technical vulnerability record.

https://nvd.nist.gov/vuln/detail/CVE-2021-44228

[26] Microsoft Security Response Center - Incident update

Microsoft's March 8, 2024 update reported access to some source repositories and internal systems following the earlier corporate-email compromise. This parent-company incident does not establish a Mojang breach.

https://www.microsoft.com/en-us/msrc/blog/2024/03/update-on-microsoft-actions-following-attack-by-nation-state-actor-midnight-blizzard

[27] Joint Task Force Transformation Initiative - Methodology

NIST SP 800-30 Rev. 1 informs assessment concepts; the 4 x 4 scores, bands, costs, owners, and residual estimates here are project-defined assumptions.

https://csrc.nist.gov/pubs/sp/800/30/r1/final

[28] OWASP Password Storage Cheat Sheet - Control guidance

Password verifiers require appropriate salted password hashing rather than reversible password encryption. Other recoverable sensitive data requires separate encryption and access controls.

https://cheatsheetseries.owasp.org/cheatsheets/Password_Storage_Cheat_Sheet.html